

Item	Description
VpSPortTrunkFlag	VpSPortTrunkFlag flag in the hardware table.
HitFlag	Whether the MAC address is matched in the hardware table. 0: no; 1: yes.

2.4.1.3 display mac-address

Function

The **display mac-address** command displays information about MAC address entries.

The **display mac-address blackhole** command displays information about static blackhole MAC address entries.

The **display mac-address static** command displays information about static MAC address entries.

The **display mac-address dynamic** command displays dynamic MAC address entries.

The **display mac-address authen** command displays information about authentication MAC address entries.

The **display mac-address total-number** command displays the number of MAC address entries of a specified type.

The **display mac-address total-number authen** command displays the number of authentication MAC address entries.

Format

For CE8855, CE8851-32CQ4BQ, CE9865, CE9855, CE8865, CE8875, CE8865-SAN, CE8851-32CQ8DQ-P, CE8850-HAM, CE8850-SAN, CE8851K, CE6866, CE6860-HAM, CE6860-SAN, CE6866K, CE6820H, CE6820H-K, CE6820S, CE6863H, CE6863H-K, CE6881H, CE6881H-K, CE6885-SAN, CE6855-48XS8CQ, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-LL (low latency mode):

```
display mac-address dynamic [ slot slot-id ] { interface { interface-type  
interface-number | interface-name } vlan vlan-id | vlan vlan-id [ interface  
{ interface-type interface-number | interface-name } ] } [ verbose ]
```

```
display mac-address mac-address [ vlan vlan-id ] [ verbose ]
```

```
display mac-address { [ vlan vlan-id ] | [ interface { interface-type interface-  
number | interface-name } ] } * [ verbose ]
```

```
display mac-address static { [ vlan vlan-id ] | [ interface { interface-type  
interface-number | interface-name } ] } * [ verbose ]
```

display mac-address **authen** { [**vlan** *vlan-id*] | [**interface** { *interface-type* *interface-number* | *interface-name* }] } * [**verbose**]

display mac-address total-number **authen** { [**vlan** *vlan-id*] | [**interface** { *interface-type* *interface-number* | *interface-name* }] } *

For CE6885-LL (low latency mode):

display mac-address total-number { [**static** | **dynamic**] { [**vlan** *vlan-id*] | [**interface** { *interface-type* *interface-number* | *interface-name* }] } * | **blackhole** [**vlan** *vlan-id*] | [**dynamic**] **slot** *slot-id* }

For CE8855, CE8851-32CQ4BQ, CE9865, CE9855, CE8865, CE8875, CE8865-SAN, CE8851-32CQ8DQ-P, CE8850-HAM, CE8850-SAN, CE8851K, CE6866, CE6860-HAM, CE6860-SAN, CE6866K, CE6863H, CE6863H-K, CE6881H, CE6881H-K, CE6885-SAN, CE6855-48XS8CQ, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L:

display mac-address total-number { [**static** | **mux** | **security** | **sticky** | **dynamic**] { [**vlan** *vlan-id*] | [**interface** { *interface-type* *interface-number* | *interface-name* }] } * | { **blackhole** | **evn** } [**vlan** *vlan-id*] | [**dynamic**] **slot** *slot-id* }

display mac-address evn [**vlan** *vlan-id*] [**verbose**]

display mac-address total-number [**static**] **bridge-domain** *bd-id*

display mac-address dynamic-blackhole [**bridge-domain** *bd-id*] [**verbose**]

display mac-address total-number dynamic-blackhole

For CE6820H, CE6820H-K, CE6820S:

display mac-address total-number { [**static** | **mux** | **security** | **sticky** | **dynamic**] { [**vlan** *vlan-id*] | [**interface** { *interface-type* *interface-number* | *interface-name* }] } * | **blackhole** [**vlan** *vlan-id*] | [**dynamic**] **slot** *slot-id* }

For CE8855, CE8851-32CQ4BQ, CE9865, CE9855, CE8865, CE8875, CE8865-SAN, CE8851-32CQ8DQ-P, CE8850-HAM, CE8850-SAN, CE8851K, CE6866, CE6860-HAM, CE6860-SAN, CE6866K, CE6863H, CE6863H-K, CE6881H, CE6881H-K, CE6885-SAN, CE6855-48XS8CQ, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-LL (low latency mode):

display mac-address blackhole [**vlan** *vlan-id*] [**verbose**]

For CE8855, CE8851-32CQ4BQ, CE9865, CE9855, CE8865, CE8875, CE8865-SAN, CE8851-32CQ8DQ-P, CE8850-HAM, CE8850-SAN, CE8851K, CE6866, CE6860-HAM, CE6860-SAN, CE6866K, CE6820H, CE6820H-K, CE6820S, CE6863H, CE6863H-K, CE6881H, CE6881H-K, CE6885-SAN, CE6855-48XS8CQ, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L:

display mac-address { **mux** | **security** | **sticky** } { [**vlan** *vlan-id*] | [**interface** { *interface-type* *interface-number* | *interface-name* }] } * [**verbose**]

Parameters

Parameter	Description	Value
<i>mac-address</i>	Displays a specified MAC address entry.	The value is in H-H-H format. An H is a hexadecimal number of 4 digits, such as 00e0 or fc01. If an H contains fewer than 4 digits, the left-most digits are padded with zeros. For example, e0 is displayed as 00e0.
vlan <i>vlan-id</i>	Displays the number of MAC address entries in a specified VLAN.	The value is an integer ranging from 1 to 4094.
verbose	Display detailed information of the static MAC address entries.	-
dynamic	Displays the number of dynamic MAC address entries.	-
interface <i>interface-name</i>	Specifies the interface name.	-
<i>interface-type</i>	Specifies the type of an interface.	-
<i>interface-number</i>	Specifies the number of an interface.	The value is a string of 1 to 47 case-sensitive characters. It cannot contain spaces.
total-number	Displays the total number of MAC address entries with multiple outbound interfaces.	-
static	Displays the number of static MAC address entries.	-

Parameter	Description	Value
mux	Displays the number of MUX MAC address entries. NOTE This parameter is supported only on the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6820H, CE6820H-K, CE6820S, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875.	-
security	Displays the number of secure-dynamic MAC address entries. This parameter is supported only by CE6820H, CE6820H-K, CE6820S, CE6863H, CE6863H-K, CE6881H and CE6881H-K. NOTE This parameter is supported only on the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6820H, CE6820H-K, CE6820S, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875.	-

Parameter	Description	Value
sticky	Displays the number of sticky MAC address entries. This parameter is supported only by CE6820H, CE6820H-K, CE6820S, CE6863H, CE6863H-K, CE6881H and CE6881H-K. NOTE This parameter is supported only on the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6820H, CE6820H-K, CE6820S, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875.	-
blackhole	Number of blackhole MAC address entries.	-
evn	Displays the number of EVN MAC address entries. This parameter is supported only on the CE6866, CE6860-HAM, CE6860-SAN, CE6866K, CE6885, CE6885L, CE6885H, CE8851-32CQ4BQ, CE8851-32CQ8DQ-P, CE8855, CE8865, CE8865-SAN, CE8875, CE9855, CE9865, CE8850-HAM, CE8850-SAN, CE8851K, CE6863H, CE6863H-K, CE6881H and CE6881H-K. NOTE This parameter is supported only on the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875.	-
slot <i>slot-id</i>	Specifies an available slot ID.	The value is a string of 1 to 23 case-sensitive characters. It cannot contain spaces.

Parameter	Description	Value
authen	Displays the number of authenticated MAC address entries.	-
bridge-domain <i>bd-id</i>	Specifies a bridge domain ID. This parameter is supported only on the CE6866, CE6860-HAM, CE6860-SAN, CE6866K, CE6885, CE6885L, CE6885H, CE8851-32CQ4BQ, CE8851-32CQ8DQ-P, CE8855, CE8865, CE8865-SAN, CE8875, CE9855, CE9865, CE8850-HAM, CE8850-SAN, CE8851K, CE6863H, CE6863H-K, CE6881H and CE6881H-K. NOTE This parameter is supported only on the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875.	For the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875: The value is an integer ranging from 1 to 16777215.
dynamic-blackhole	Displays dynamic blackhole entries. This parameter is supported only on the CE6866, CE6860-HAM, CE6860-SAN, CE6866K, CE6885, CE6885L, CE6885H, CE8851-32CQ4BQ, CE8851-32CQ8DQ-P, CE8855, CE8865, CE8865-SAN, CE8875, CE9855, CE9865, CE8850-HAM, CE8850-SAN, CE8851K, CE6863H, CE6863H-K, CE6881H and CE6881H-K. NOTE This parameter is supported only on the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875.	-

Views

All views

Default Level

1: Monitoring level

Usage Guidelines

Usage Scenario

A MAC address table is a port-based Layer 2 forwarding table. It stores MAC addresses learned by a device from other devices, including MAC addresses, ports associated with the MAC addresses, and IDs of VLANs, BDs, and VSIs to which the specified ports belong. When forwarding data, the device searches the MAC address table for the outbound interface based on the destination MAC address in the packet. This reduces broadcast traffic.

MAC address entries are classified into the following types:

- Static MAC address entries: They are manually configured so that packets with specified destination MAC addresses are forwarded by specified interfaces. Static MAC address entries protect devices against attacks with forged MAC addresses. Static MAC address entries do not age.
- Dynamic MAC address entries: obtained by learning source MAC addresses. Dynamic MAC address entries can be aged out after the aging time expires.
- Static blackhole MAC address entries: manually configured by users. To prevent useless MAC address entries (such as MAC address entries of unauthorized users) from occupying the MAC address table and prevent hackers from using MAC addresses to attack user devices or networks, you can configure MAC addresses of untrusted users as static blackhole MAC addresses. Packets destined for these untrusted MAC addresses are discarded. These entries will not be aged out.

Prerequisites

A static blackhole MAC address entry has been configured using the **mac-address blackhole** command in the system view, a static MAC address entry has been configured using the **mac-address static vlan** command, or a MAC address entry has been configured using another command.

Precautions

If a device has a large number of MAC address entries, it is recommended that you specify a VLAN or VSI in the **display mac-address** command to filter the output information. Otherwise, excessive output information may cause the following problems:

- The terminal screen is refreshed continuously and the required information cannot be obtained.
- The system traverses and retrieves information for a long time. As a result, the system does not respond.

If a large number of static MAC address entries are configured on a device, it is recommended that you specify a VLAN or VSI in the **display mac-address static** command to filter the output information. Otherwise, excessive output information may cause the following problems:

- The terminal screen is refreshed continuously and the required information cannot be obtained.

- The system traverses and retrieves information for a long time. As a result, the system does not respond.

Example

NOTE

The actual command output varies according to the device. The command output here is only an example.

Display the number of dynamic MAC addresses in VLAN 100.

```
<HUAWEI> display mac-address total-number dynamic vlan 100
Total number of mac-address : 50
```

Display all static blackhole MAC address entries.

```
<HUAWEI> display mac-address blackhole
---- Flags: * - Backup
      BD: bridge-domain      Age: dynamic MAC learned time in
seconds
-----
MAC Address      VLAN/VSI/BD      Learned-From      Type
Age
-----
00-e0-fc-12-34-56 200/-/-          -                  blackhole        -
-----
Total items: 1
```

Display detailed information about static MAC address entries.

```
<HUAWEI> display mac-address static verbose
BD: bridge-domain      Age: dynamic MAC learned time in seconds
MAC address table:
-----
MAC Address: 00-e0-fc-12-34-56      VLAN/VSI/BD      : 10/-/-
Type      : static                  Learned-From      : 100GE1/0/1
PEVLAN    : 10                     CEVLAN           : -
Aging time : -
-----
Total items displayed = 1
```

Display the detailed information about static blackhole MAC address entries.

```
<HUAWEI> display mac-address blackhole verbose
BD: bridge-domain      Age: dynamic MAC learned time in seconds
MAC address table:
-----
MAC Address: 00-e0-fc-12-34-56      VLAN/VSI/BD      : 10/-/-
Type      : blackhole                Learned-From      : -
PEVLAN    : 10                     CEVLAN           : -
Aging time : -
-----
Total items displayed = 1
```

Display the number of static MAC address entries.

```
<HUAWEI> display mac-address total-number static
Total number of mac-address : 600
```

Display the number of dynamic MAC address entries on 100GE 1/0/1.

```
<HUAWEI> display mac-address total-number dynamic interface 100ge 1/0/1
Total number of mac-address : 5000
```

Table 2-240 Description of the **display mac-address** command output

Item	Description
Total number of mac-address	Number of MAC address entries in the system.
Total items	Total number of MAC address entries that match specific rules.
MAC Address	Configured static MAC address, MAC address in a specific entry, or configured static blackhole MAC address.
MAC address table	MAC address table.
VLAN/VSI/BD	• VLAN: ID of the VLAN to which the outbound interface belongs. • VSI: name of the VSI to which the outbound interface belongs. • BD: ID of the BD to which the outbound interface belongs.
Learned-From	• Static MAC address configured on this interface when the MAC address type is static. • MAC address learned on this interface when the MAC address type is dynamic. • A hyphen (-) is displayed when the MAC address type is blackhole.
Type	Type of a MAC address. • static: Static MAC address entries which are configured by users. They will not age out. They remain in the MAC address table after the system is restarted or a board is hot swapped. • blackhole: Static blackhole MAC address entries which are configured by users. They will not age out. They remain in the MAC address table after the system is restarted or a board is hot swapped. • dynamic: Dynamic MAC address entries which are obtained by learning source MAC addresses. They will age out after the aging time expires. They are removed from the MAC address table after the system is restarted or a board is hot swapped.
Age	Dynamic MAC learned time in seconds.

Item	Description
PEVLAN	Single or outer VLAN tag carried in the packets received by the interface.
CEVLAN	Inner VLAN tag in the double VLAN tags of packets received by an interface.
Aging time	Aging time of MAC address entries.
BD	BD ID.

2.4.1.4 display mac-address aging-time

Function

The **display mac-address aging-time** command displays the aging time of MAC address entries.

Format

display mac-address aging-time

Parameters

None

Views

All views

Default Level

1: Monitoring level

Usage Guidelines

Usage Scenario

After the aging time of MAC address entries is configured using the **mac-address aging-time** command, to check the aging time, run the **display mac-address aging-time** command.

Example

NOTE

The actual command output varies according to the device. The command output here is only an example.

```
# Display the aging time of MAC address entries.  
<HUAWEI> display mac-address aging-time  
Aging time: 300 second(s)
```